

Flight Plan Transformer

User Guide

ForeFlight to Garmin GPSMAP 696, on iPhone and iPad

Version 2026.2.0 · October 2026

Questions or suggestions: **eric@thueson.com**

Contents

1. What the app does
2. What you need
3. Install the app (one time)
4. Set up the app and folders (one time)
5. Export a flight plan from ForeFlight
6. Open and convert the flight plan
7. Check the waypoint names
8. Add leg altitudes (optional)
9. Save the file to the SD card
10. Import into the Garmin 696
11. Business rules
12. Updates
13. Troubleshooting
14. Quick reference

1. What the app does

ForeFlight can export a flight plan as a Garmin `.fpl` file, but the Garmin GPSMAP 696 will not read that file as it is. Flight Plan Transformer converts it on your iPhone or iPad so the 696 can import it from an SD card.

The app follows a list of rules: it removes information the 696 does not use, adds fields it needs, rounds coordinates, and gives map points names the 696 accepts. You can see every change before you save.

> **Tip:** when you save, type a **Plan name on the GPS** such as `KDIJ LOOP`. That is the name the Garmin 696 shows in its flight plan list (section 9).

The app runs on your device with no internet connection once installed. Your flight plans never leave your iPhone or iPad.

2. What you need

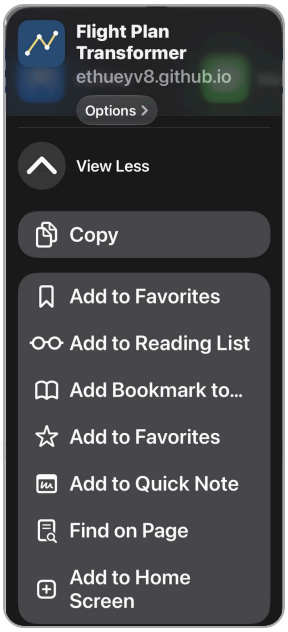
- An iPhone or iPad with Safari
- A ForeFlight account, and knowing how to build a route in it
- An SD card reader that plugs into your iPhone or iPad (Lightning or USB-C)

- The SD card used with the Garmin GPSMAP 696
- Internet access for the one-time install

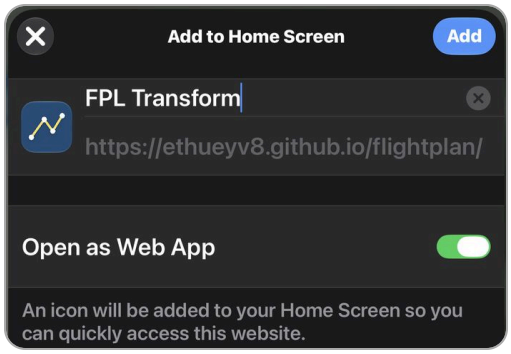
3. Install the app (one time)

The app is a web app. It installs from Safari onto your home screen and then works like any other app. It does not come from the App Store.

1. On your iPhone or iPad, open **Safari** and go to `https://ethueyv8.github.io/flightplan/`
2. Tap the **Share** button.
3. Tap **Add to Home Screen** (tap **View More** if you do not see it).
4. Name it **FPL Transform**, leave **Open as Web App** on, and tap **Add**.



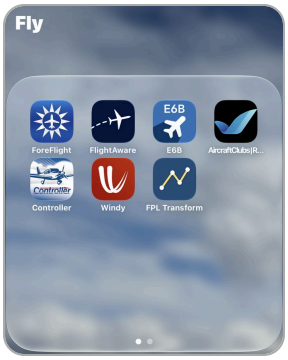
Safari Share menu



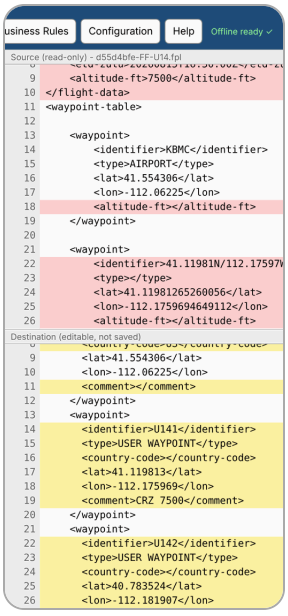
Add to Home Screen

5. Close Safari. Open the app from the new **FPL Transform** icon on your home screen, while you are still online.
6. Swipe the blue button bar to the left to see its end. When it shows **Offline ready ✓**, the app works without internet. If it says *Not offline ready*, close the app and open it again from the icon.
7. Test it: turn on airplane mode and open the app from the icon.

> Always open the app from the home screen icon, not from Safari. The icon keeps its own offline copy.



The FPL Transform icon

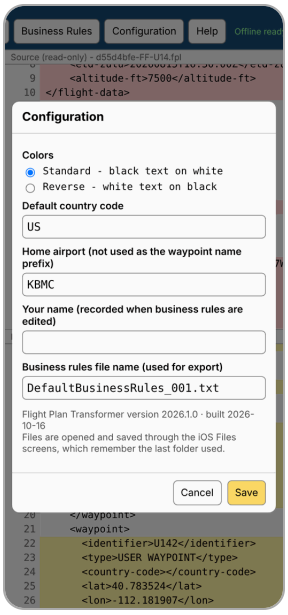


End of the button bar

4. Set up the app and folders (one time)

App settings

Tap **Configuration** (swipe the button bar to find it) and check these settings, then tap **Save**.



Configuration

Setting	Set to
Colors	Standard (black on white) or Reverse (white on black)
Default country code	`US`
Home airport	Your home field, for example `KBMC`. It is never used to name map points.
Your waypoint letter	One letter, different for each pilot, for example `E` and `V`. New waypoint names start with it. `K` is not allowed because names like `KDIJ1` look like airport codes.
Waypoint name register	Shows how many names this device remembers. **Load plans to check** adds names from saved plans or a GPS waypoint export. **Clear register** forgets them.
Your name	Recorded when you change the business rules
Business rules file name	Leave as is

The bottom of the screen shows the app version and build date.

Folders for your flight plans

Keep the files from ForeFlight apart from the converted files. In the **Files** app, tap **Browse**, then **iCloud Drive**:

1. Create a folder named **Flightplans** (tap the New Folder button, the folder with a +).
2. Open it and create a folder inside it named **ForeFlight Flightplans**.

Folder	Holds
iCloud Drive / Flightplans / ForeFlight Flightplans	Files exported from ForeFlight (the source files)
iCloud Drive / Flightplans	Converted files ready for the Garmin

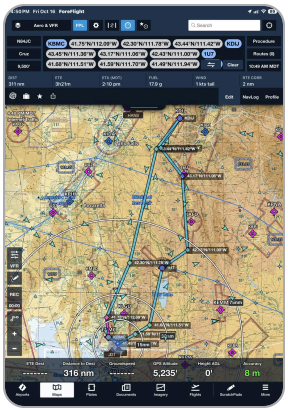
> Create the folders in **iCloud Drive**, not *On My iPhone* or *On My iPad*. Folders on the device can only be seen on that device. iCloud Drive folders appear on your iPhone, iPad and computer.

5. Export a flight plan from ForeFlight

> **Use the ForeFlight app on your iPhone or iPad.** ForeFlight in a computer web browser cannot save a flight plan file. Export the plan from the ForeFlight app on the same iPhone or iPad you use for this app (or another device signed in to the same iCloud account).

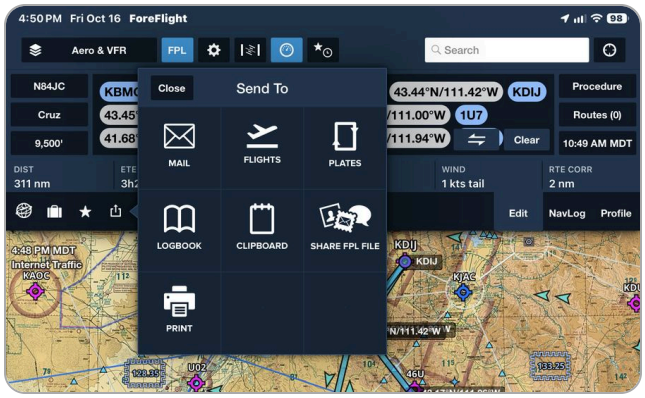
1. In ForeFlight, open **Maps** and build your route as usual. The route shows in the flight plan bar (**FPL**) at the top.
2. Tap the **share** icon (a box with an arrow) on the toolbar below the route.

> Points you tap on the map are named after their coordinates, such as `41.75°N/112.09°W`, and ForeFlight does not let you rename them. The 696 cannot use those names and silently drops those points, so the app renames them for you (section 7).

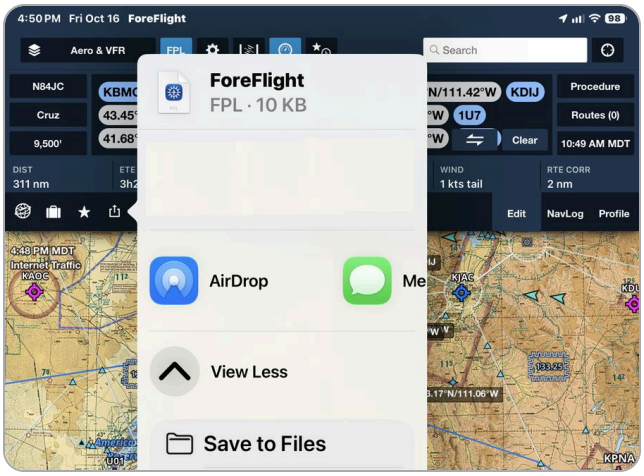


Route in ForeFlight

3. The **Send To** menu opens. Tap **Share FPL File**.
4. Tap **Save to Files**. If it is not listed, tap **View More**.

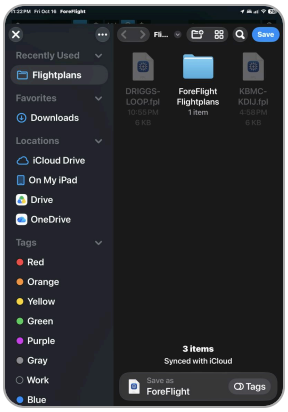


Send To → Share FPL File



Save to Files

5. Go to **iCloud Drive → Flightplans → ForeFlight Flightplans**.
6. Give the file a name if you like, for example the route, and tap **Save**. The app does not use this name.



Save to Files from ForeFlight. Open the ForeFlight Flightplans folder, then tap Save.

6. Open and convert the flight plan

Open the ForeFlight file

1. Open the **FPL Transform** app from its home screen icon.
2. Tap **Open Source**, go to **iCloud Drive → Flightplans → ForeFlight Flightplans**, and pick the file you saved from ForeFlight.

The file appears in the top pane, called the **source**. It is read-only; the app never changes your original file.



Source file opened (iPad)

Apply the rules

Tap **Apply Rules**. The converted file, called the **destination**, appears in the bottom pane.



After Apply Rules (iPad)

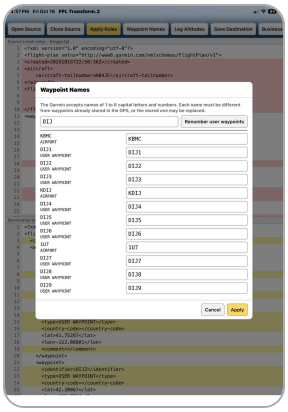
- **Yellow** lines in the bottom pane were added or changed.
- **Red** lines in the top pane were removed.
- Line numbers are on the left. Long lines do not wrap.
- The two panes scroll together, up and down and left and right, so matching lines stay side by side.
- You can type in the bottom pane. Your edits are highlighted too.

If a message lists warnings, read them. They name anything the 696 may not accept.

What the rules change

ForeFlight file	After conversion
Date `20261016T04:04:10Z`	`2026-10-16T04:04:10Z`
Aircraft and departure-time blocks	Removed
Map points named `41.11981N/112.17597W` with a blank type	Renamed (for example `U141`), type `USER WAYPOINT`
No country code	`US` for airports and fixes, blank for user waypoints
Coordinates with up to 16 decimals	Rounded to 6 decimals (within 4 inches)
Waypoint altitude field	Removed; an empty comment is added for altitude notes
Route name `KBMC TO U14`	`KBMC/U14`
Flight plan number	Removed
Layout	Indented and spaced like files the 696 writes itself

7. Check the waypoint names



Waypoint Names (iPad)

Waypoint names must be **capital letters or numbers only**. The 696 stores longer names, but its flight plan page shows only the first 8 characters, so the app keeps names to **8 or fewer**. The app renames every point that does not fit, using a short prefix and a number.

The prefix is the first airport in the route that is not your home airport, without its leading K. For the KBMC → KDIJ → 1U7 → KBMC loop shown here, the points become `DIJ1`, `DIJ2`, and so on. Names that are already valid, such as `EMONT`, are kept.

Tap **Waypoint Names** to see every name:

- Type over a name to change it.
- Or type a new prefix at the top and tap **Renumber user waypoints**.
- Tap **Apply**. The app refuses invalid or duplicate names.

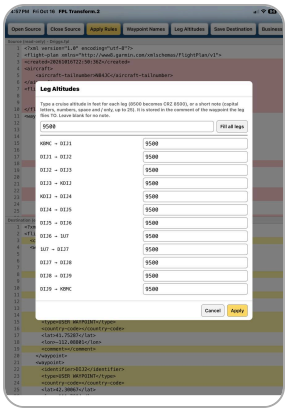
Two pilots, one GPS: keeping names unique. The 696 keeps every user waypoint it imports. If a name is already stored at a different place, the 696 does not move the old point; it saves the new one with a `0` added (`DIJ1` becomes `DIJ10`), so the names on the GPS no longer match your plan and look alike. The app avoids this:

- Each pilot sets a different **Your waypoint letter** (section 4). New names start with it: `EDIJ1`, `VDIJ1`.
- The app remembers every user waypoint name it saves, with its position. Numbering skips names already used elsewhere. A point that is already stored at the same position, under any name (for example `DIJ1` from the GPS export), keeps that name, so no duplicate waypoint is created.
- When you save, the app **refuses a name that is already used at another place** and offers **Rename these automatically**.
- The memory is per device. To include names made on your other device or by your partner, open **Configuration → Load plans to check** and select their saved `.fpl` files (for example the whole Flightplans folder), or the 696's waypoint export (below).
- **Best check:** on the 696, export the user waypoints to the SD card (User Waypoints, then **MENU**, **Export**; the file is `Waypoints1.gpx`) and load that file. It holds every name in the GPS, from both pilots.

After every import, compare the 696's total distance with ForeFlight's. A misplaced waypoint changes it.

8. Add leg altitudes (optional)

ForeFlight sets one altitude for the whole flight, and the 696 has no altitude for each leg. The app can store a planned altitude as a note in each waypoint's comment, which you can read on the 696.



Leg Altitudes (iPad)

1. Tap **Leg Altitudes**.
2. Type an altitude in feet for each leg. `8500` becomes `CRZ 8500`. A short note also works, such as `DESCEND 6500`.
3. Or tap **Fill all legs** to use the altitude from ForeFlight.
4. Tap **Apply**.

The note goes on the waypoint the leg flies *to*. Notes may use capital letters, numbers, spaces and `/`, up to 25 characters.

Reading the altitude on the 696

The flight plan page does not show the notes. To see one:



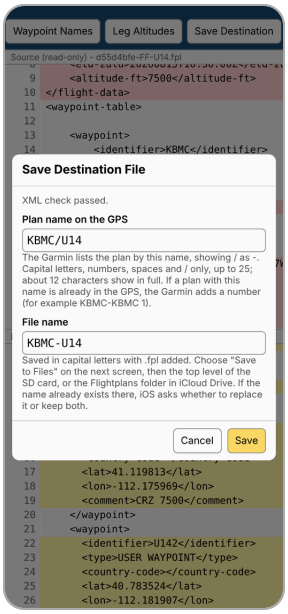
User Waypoint page: CRZ 9500 in the INFO box

1. Open the flight plan on the 696.
2. Press the **down** arrow to highlight a user waypoint, for example `DIJ2`.
3. Press **ENT**. The **User Waypoint** page opens. The note, such as `CRZ 9500`, is at the top right of the **INFO** box, above the coordinates.
4. Press **ENT** again on **Done** to close it.

> Only user waypoints (the renamed map points) keep a note. Airports and named fixes come from the Garmin database and do not show one, so a leg that ends at an airport has no altitude note on the 696.

9. Save the converted file

1. Tap **Save Destination**. The app checks the file. *XML check passed* means it is well formed. If the check fails, see Troubleshooting.
2. **Plan name on the GPS:** the 696 lists the plan by this name, not by the file name, and shows `/` as `-`. It starts as the route, such as `KBMC/U14`. Give round trips a clear name such as `DRIGGS`, because every KBMC round trip would otherwise be called `KBMC-KBMC`. Use capital letters, numbers, spaces and `/`; about 12 characters show in full.
3. Check the file name. It follows the plan name, in capital letters with `.fpl` added.
4. Tap **Save**, then **Save to Files**.
5. Choose where to save, using option A or B below, and tap **Save**. If a file with the same name is already there, iOS asks whether to replace it or keep both.



Save Destination

> **Name your plan.** The 696 lists each plan by the **Plan name on the GPS**, not by the file name. Without a name, every round trip from home appears as `KBMC-KBMC`, and a second one becomes `KBMC-KBMC 1`. Type a short, clear name such as `KDIJ LOOP` each time you save.

Option A: straight to the SD card

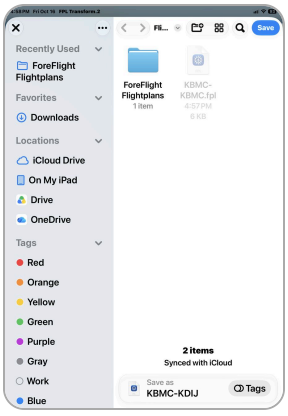
Plug the SD card reader, with the card in it, into your iPhone or iPad. In Save to Files, choose the SD card under **Locations** and save at the **top level** of the card, not inside a folder.

Option B: to iCloud Drive, then copy to the card

Save in **iCloud Drive → Flightplans**. The file is then on all your devices, and you can copy it to the card later:

1. Plug in the SD card reader and open the **Files** app.
2. Go to **iCloud Drive → Flightplans**. Touch and hold the file, then tap **Copy**.
3. Under **Locations**, open the SD card. Touch and hold an empty spot at the top level, then tap **Paste**.

When you are done, unplug the SD card reader.



Saving in iCloud Drive / Flightplans

10. Import into the Garmin 696

1. Take the card out of the reader and insert it into the 696.
2. Press the **FPL** key, then turn the FMS joystick to show the **Flight Plan List** page.



Flight Plan List. DIJ-BMC was named in the app.

3. Press the **IMPORT** softkey. The plans on the SD card are listed.
4. Highlight your plan with the FMS joystick.
5. Press **IMPORT**, then **ENT**.

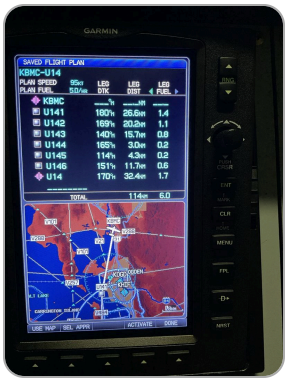


Flight Plans on Card

6. Open the imported plan, **count the waypoints** and **compare the total distance** with ForeFlight. The 696 says the import succeeded even when it skips points it cannot read.
7. Check the route on the map.

> **Always check the imported route on the 696 before flight.** The app helps prepare the file; the pilot is responsible for the route.

How the 696 names imported plans



Imported plan: waypoints, legs and map

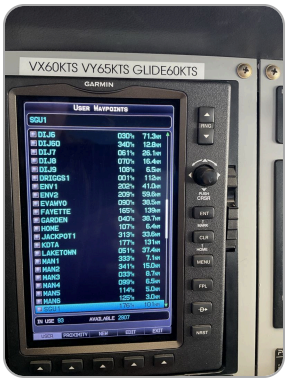
- The plan is listed by its **plan name** (section 9), not by the file name.
- If a plan with the same name is already stored, the 696 adds a number, for example `KBMC-KBMC 1`. Delete the old plan first, or use a different plan name.
- The 696 holds 50 plans; the list shows how many are in use and available.
- **Deleting a plan does not delete its user waypoints.** Remove ones you no longer need from the user waypoint list.
- If a user waypoint name is already stored at a **different place**, the 696 keeps the stored one and saves the new point with a `0` added (`DIJ1` becomes `DIJ10`). Older plans are not changed, but the names in the new plan no longer match the app.
- The User Waypoints list shows full names; the flight plan page shows only the first 8 characters.
- When you delete a waypoint that a plan still uses, the 696 warns *It is used in a flight plan*. Choose **No** unless you no longer need that plan.

Finding the stored user waypoints

1. From the main map screen, press **MENU**, then **MENU** again to open the **Main Menu**.
2. Turn the FMS joystick to **User Waypoints** and press **ENT**.
3. The list shows every user waypoint stored in the 696, with how many are in use. Highlight one and press **ENT** to see its page, including any altitude note, or to delete it.



Main Menu



User Waypoints list

Check this list before reusing a prefix, and delete leftovers from old or test plans.

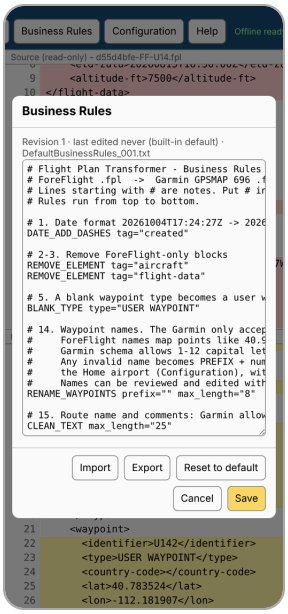
11. Business rules

The business rules are the list of conversion steps the app follows, one per line. Lines starting with `#` are notes. Put `#` in front of a rule to switch it off. You do not need to change them for normal use.

Tap **Business Rules** to see them. The top shows the revision, when they were last changed, and by whom.

- **Save** keeps your changes on this device.
- **Export** saves the rules as a text file, to keep or share with a partner.
- **Import** loads a rules file someone shared with you. Tap Save to keep it.
- **Reset to default** brings back the built-in rules.

> Rules are stored on this device only. Deleting the app icon erases them, so export a copy if you change them.



Business Rules

12. Updates

The version number is the year followed by the release, for example `2026.2.0`. It appears in Configuration and at the top of Help.

1. While online, open the app from the home screen icon, close it completely, and open it again.
2. Check the version in Configuration.
3. If it has not changed: export your business rules if you changed them, delete the icon, then install again as in section 3.

13. Troubleshooting

Problem	What to do
Airports appear on the 696 but some waypoints are missing	A waypoint name is not valid. Open **Waypoint Names**; each name must be 1 to 8 capital letters or numbers. Check the warnings after Apply Rules.
696 says the import succeeded but the plan is empty or missing	Check that the file was saved at the top level of the card with the `.fpl` ending, and that there were no warnings.
The plan is not in the 696 import list	Save the file at the top level of the SD card, not in a folder.
Plan shows with a number added (KBMC-KBMC 1)	A plan with that name was already stored. Use a different plan name when saving, or delete the old plan on the 696.
Old waypoints still show on the map after deleting a plan	Deleting a plan keeps its user waypoints. Delete them in MENU, MENU, User Waypoints.
A stored waypoint moved, or an older plan changed	Two plans used the same waypoint name. Renumber with a different prefix.
XML check FAILED when saving	An edit in the bottom pane broke a tag. Tap **Apply Rules** to start the conversion again.
The ForeFlight file is greyed out in Open Source	Check the file was saved from ForeFlight to Files and ends in `.fpl`.
The app will not open in airplane mode	Open it from the home screen icon while online until the bar shows Offline ready.
Results look wrong after an update	Business Rules, then **Reset to default**, then Save.
The folders are missing on another device	They were created *On My iPhone/iPad*. Create them in **iCloud Drive** and move the files there.
The SD card does not appear in Save to Files	Unplug and reconnect the reader. In Files, tap Browse and look under Locations.

14. Quick reference

1	**ForeFlight app on iPhone/iPad** (not a computer browser): share icon, Share FPL File, Save to Files, *Flightplans / ForeFlight Flightplans*.
2	**App:** Open Source, pick the file.
3	**Apply Rules.** Read any warnings.
4	**Waypoint Names:** check the names; use a new prefix if a name was used in another plan.
5	**Leg Altitudes** (optional).
6	**Save Destination:** set a clear plan name; Save to Files, top level of the SD card; or *Flightplans* in iCloud Drive, then copy to the card.
7	**Garmin 696:** FPL key, Flight Plan List, IMPORT, select, IMPORT, ENT.
8	**Count the waypoints** and check the route on the map before flight.

Flight Plan Transformer version 2026.2.0. App address: https://ethueyv8.github.io/flightplan/